Curatio BioBERT Triage
Methodology and Results

From patient complaint to colour-coded clinical pathways

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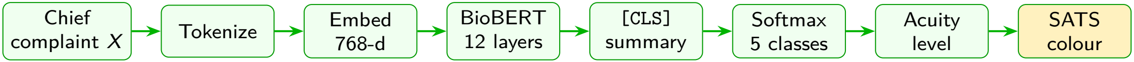
Outline

How input flows through the model



Input text passes through these stages to produce a **colour-coded triage decision**.

Stage 1: Tokenization

Computers read numbers, not words. Tokenization maps text to integer IDs:

$$X \xrightarrow{\mathcal{T}} (t_1, t_2, \ldots, t_M), \qquad M \leq 128$$

- **WordPiece:** common words stay whole; rare words split ("headache" → `head+##ache`)
- Vocabulary $\mathcal{V}$: $|\mathcal{V}| = 28{,}996$ token IDs
- Special tokens: `[CLS]` (101), `[SEP]` (102), `<redacted_PAD>` (0)

i	Token	ID
0	[CLS]	101
1	i	1045
2	have	2031
4	head+##ache	7994
7	fever+##ish	9643
9	[SEP]	102
10–127	<redacted_PAD>	0

Stage 2: Embedding

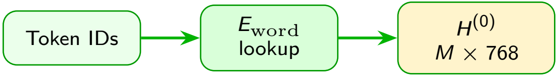
Each token ID is converted to a dense vector by table lookup:

$$\mathbf{e}_i = E_{\text{word}}[\text{id}(t_i)] \in \mathbb{R}^{768}$$

Three learned vectors are summed for each position:

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(t_i)$$

- E_{word}: what word is this?
- $E_{\text{pos}}(i)$: where is it in the sentence?
- E_{seg}: which sentence segment



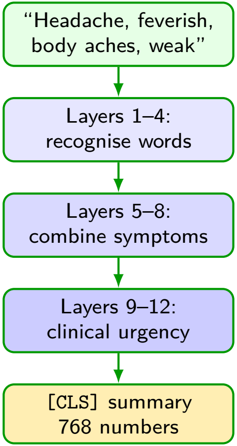
All token vectors form:

$$H^{(0)} \in \mathbb{R}^{M \times 768}$$

Simple meaning

Row 0 of $H^{(0)}$ is `[CLS]` — this row becomes the triage summary after BioBERT.

Stage 3: BioBERT processing



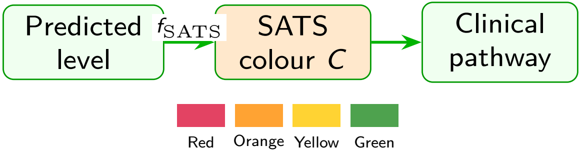
What BioBERT does (in words):

- Reads the complaint **in both directions** at once
- Each of 12 layers lets every word "look at" every other word
- Clinical words like *headache* and *feverish* receive more weight
- After 12 layers, the `[CLS]` token holds a compressed summary of the whole complaint
- This summary is passed to a classifier that picks the acuity level

Stage 4: Acuity level to SATS colour

Level → colour mapping

Level	Colour	Target time
1	Red	Immediate
2	Orange	< 10 min
3	Yellow	< 60 min
4–5	Green	< 4 hr



Each colour triggers a **clinical pathway** — the ED routes the patient to the right department.

Simple meaning

The model predicts a number (1–5); the hospital system converts it to a colour and routes the patient.

Project goal

Research question: Can NLP map an unstructured chief complaint to the correct acuity level and colour-coded clinical pathway?

Formal objective

$$\text{Input } X \rightarrow \hat{c} \in \{1, \ldots, 5\} \rightarrow C = f_{\mathrm{SATS}}(\hat{c})$$

What it does: converts patient text into an acuity level, then into a SATS colour for routing.

Symbol	Meaning	Role in triage
X	Chief complaint text	Raw patient input (max 128 tokens)
$\hat{c}$	Predicted acuity level	Model output before colour mapping
C	SATS colour	Red / Orange / Yellow / Green pathway
f_{SATS}	Level-to-colour map	Fixed rule: 1 $\rightarrow$Red, 2 $\rightarrow$Orange, 3$\rightarrow$Yellow, 4, 5$\rightarrow$Green

Was the goal achieved?

Answer: Yes — on 8,000-sample holdout

99.92% of complaints were routed to the correct SATS colour (baseline BioBERT).

- 7,994 of 8,000 complaints: predicted colour matches nurse label
- 6 errors on Level 1 (Red) — the most critical safety class
- Holdout = 10% stratified split the model did not train on

Simple meaning

Because f_{SATS} is a fixed mapping, getting the acuity level right means getting the colour pathway right. Next slide defines the accuracy formula and every symbol.

Variables: colour-routing accuracy

What it computes: the fraction of patients routed to the correct colour.

$$P(f_{\mathrm{SATS}}(\hat{c}) = f_{\mathrm{SATS}}(c_{\mathrm{nurse}})) = 0.9992$$

$$\mathrm{Accuracy} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}[\hat{c}_i = c_i] = 0.9992$$

Symbol	Meaning	Role in triage
$\hat{c}_i$	Model's predicted level for patient i	BioBERT output before colour map
c_i	Nurse's true acuity label	Ground truth from training data
c_{nurse}	Same as c_i (one patient)	Reference label to compare against
n	Number of test complaints	$n = 8{,}000$ on holdout
$\mathbb{1}[\cdot]$	Indicator function	$= 1$ if prediction correct, $= 0$ if wrong
f_{SATS}	Level $\rightarrow$ colour function	Turns level into routing colour

Result: $\frac{7994}{8000} = 0.9992$.

How we measure success

We tested on **8,000 complaints** the model had not seen during training (10% stratified holdout).

Metric	**Simple formula**
Accuracy	$\frac{\text{correct}}{\text{total}} = \frac{7994}{8000}$
Precision	$\frac{TP}{TP+FP}$
Recall	$\frac{TP}{TP+FN}$
F1	$2 \cdot \frac{\text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}}$
Macro-F1	average F1 across all 5 levels

What each one measures:

- **Accuracy** — overall correct routing rate
- **Precision** — when we say “Red”, how often correct?
- **Recall** — of all real Reds, how many caught?
- **F1** — balance of precision and recall
- **Macro-F1** — each level weighted equally

Worked example

Next slide defines TP, FP, FN and how they enter precision and recall.

Variables: TP, FP, FN

What they count (for one acuity class, e.g. Level 1 Red):

Symbol	**Meaning**	**Role in triage**
TP	True positives	Correctly predicted as this class
FP	False positives	Wrongly predicted as this class
FN	False negatives	This class missed by the model

Example (L1 Red): 322 true Reds; model correctly flags 316 (TP) and misses 6 (FN); no false Reds ($FP = 0$).

Variables: Precision, Recall, Macro-F1

What the formulas compute: per-class quality and an average across all five levels.

Symbol	**Meaning**	**Role in triage**
Prec	$TP/(TP+FP)$	When we predict this class, how often right?
Rec	$TP/(TP+FN)$	Of all true cases, how many did we catch?
$F1$	$2 \cdot \frac{\text{Prec}\cdot\text{Rec}}{\text{Prec}+\text{Rec}}$	Single score balancing both
Macro-F1	$\frac{1}{5}\sum_{c=1}^{5} F1_c$	Each level weighted equally

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \qquad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}$$

Simple meaning

Macro-F1 averages F1 across all 5 levels so minority classes (Red) count as much as majority (Yellow).

Model performance: baseline vs SMOTE

Default inference model: baseline BioBERT.

Metric	Baseline	SMOTE
Accuracy	99.92%	99.85%
Macro-F1	99.77%	99.54%
Recall L1 (Red)	98.14%	100.00%
Mean confidence $\bar{c}$	0.9997	0.9992
Predictions with $\hat{y}_{\hat{c}} = 1.0$	79.05%	82.99%
Latency p50 (ms/sample)	100.3	95.5

Symbol	Meaning	Role in triage
$\bar{c}$	Mean softmax confidence	Average of top-class probabilities
$\hat{y}_{\hat{c}}$	Confidence of predicted class	How sure the model is about its pick
SMOTE	Oversampled training variant	Alternative model for minority-class recall

Worked example: L1 Red recall

What it computes: of all true Red patients, what fraction did we correctly route to Red?

$$\mathrm{Rec}_{\mathrm{L1}} = \frac{TP_{\mathrm{L1}}}{TP_{\mathrm{L1}} + FN_{\mathrm{L1}}} = \frac{316}{322} = 0.9814$$

Symbol	Meaning	Role in triage
TP_{L1}	True Red cases correctly predicted Red	316 patients
FN_{L1}	True Red cases missed (wrong colour)	6 patients
322	Total true Red in holdout	Support count for Level 1
$\mathrm{Rec}_{\mathrm{L1}}$	L1 recall	Safety metric: missed emergencies

Simple meaning

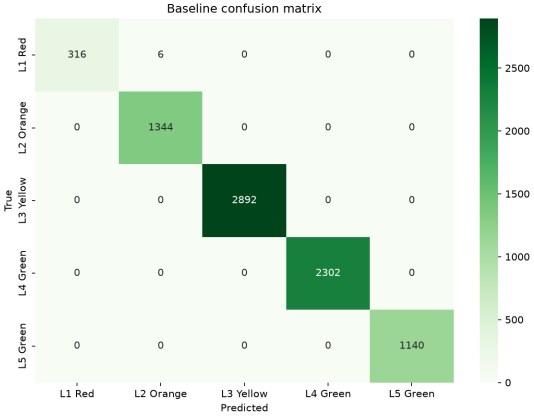
Six Red patients were assigned a less urgent colour — the main clinical risk on this holdout.

Per-class results and confusion matrix

Baseline per-class

Class	Prec	Rec	F1	*n*
L1 Red	1.00	0.98	0.99	322
L2 Orange	1.00	1.00	1.00	1344
L3 Yellow	1.00	1.00	1.00	2892
L4 Green	1.00	1.00	1.00	2302
L5 Green	1.00	1.00	1.00	1140

Confusion matrix: $\sum_i C_{ii} = 7{,}994$ correct; 6 off-diagonal errors.



Variables: confusion matrix

What it shows: how many patients with true level i were predicted as level j.

Symbol	**Meaning**	**Role in triage**
C_{ij}	Count: true level i, predicted j	Cell in confusion matrix
C_{ii}	Diagonal entries	Correct routing (same level)
i, j	Class indices $1, \ldots, 5$	Rows $=$ truth, columns $=$ prediction
n	Row sum $\sum_j C_{ij}$	Number of true cases per class

Result: $\sum_i C_{ii} = 7{,}994$ correct; all 6 errors involve true Level 1 (Red).

Variables: model confidence

What it computes: how strongly the softmax favours the predicted class.

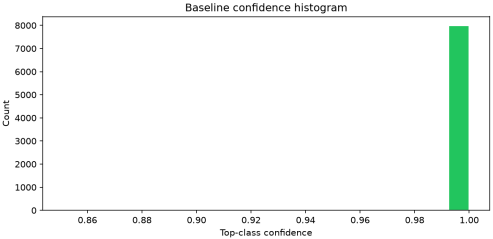
$$\hat{y}_{\hat{c}} = \max_{c\in\{1,\dots,5\}} \hat{y}_c, \qquad \bar{c} = \frac{1}{n}\sum_{i=1}^{n} \hat{y}_{\hat{c}_i}$$

Symbol	**Meaning**	**Role in triage**
$\hat{y}_c$	Softmax probability for level c	Model belief for each class; $\sum_c \hat{y}_c = 1$
$\hat{c}$	$\arg\max_c \hat{y}_c$	Predicted acuity level (highest probability)
$\hat{y}_{\hat{c}}$	Probability of the predicted class	Confidence score shown in the app
$\bar{c}$	Mean confidence over n samples	Average certainty on holdout ($=$ 0.9997)
n	Holdout size	$n = 8{,}000$ complaints

Result: 79.05% of predictions have $\hat{y}_{\hat{c}} = 1.0$ exactly.

Calibration results

What calibration means: when the model says 90% confident, it should be right ∼90% of the time. Our model is **overconfident**.



- $\bar{c} = 0.9997$ — very high mean confidence
- $P(\hat{y}_{\hat{c}} = 1.0) = 0.7905$ — 79% at maximum
- $P(\hat{y}_{\hat{c}} < \tau) = 0$ where $\tau = 0.85$

Variables: calibration threshold

What these symbols mean for deployment safety:

Symbol	Meaning	Role in triage
τ	Confidence threshold	Below τ, flag for Bayesian / manual review
$P(\cdot)$	Probability over holdout	Fraction of predictions meeting a condition
Calibration	Confidence vs correctness	Miscalibration = trusting softmax too much

Simple meaning

Cross-entropy training rewards confident correct answers, so probabilities cluster near 1.0. TEWS vitals remain essential for safe triage.

Goal vs evidence

Goal

- Predict acuity level from text
- Route patient to correct SATS colour
- Support colour-coded clinical pathways

Evidence (baseline)

- Accuracy: **99.92%**
- Macro-F1: **99.77%**
- L1 (Red) recall: **98.14%**
- Colour routing: **7,994 / 8,000** correct

Limitations:

- Holdout is in-distribution — live KATH data may differ
- 79% of predictions overconfident ($\hat{y}_{\hat{c}} = 1.0$)
- Text-only — vitals (TEWS) needed for full clinical safety

Conclusion

Pipeline recap: complaint $\rightarrow$ tokenize $\rightarrow$ embed $\rightarrow$ BioBERT $\rightarrow$ acuity level $\rightarrow$ SATS colour $\rightarrow$ clinical pathway.

1. The goal — **acuity prediction and colour-coded routing** — was achieved at 99.92% on holdout
2. Baseline BioBERT is the default model (99.77% macro-F1)
3. The open mathematical problem is **calibration**: softmax overconfidence means TEWS vitals and Bayesian fusion remain essential safety layers

Questions?